

MNIST Handwritten Digit Classification

Project Overview

A deep learning project designed to recognize handwritten digits from 0-9. It implements a neural network to process 28x28 grayscale images with high precision.

Technology Stack



Core Engine

Python 3.x utilizing NumPy for numerical computation and Pandas for data handling.



Deep Learning

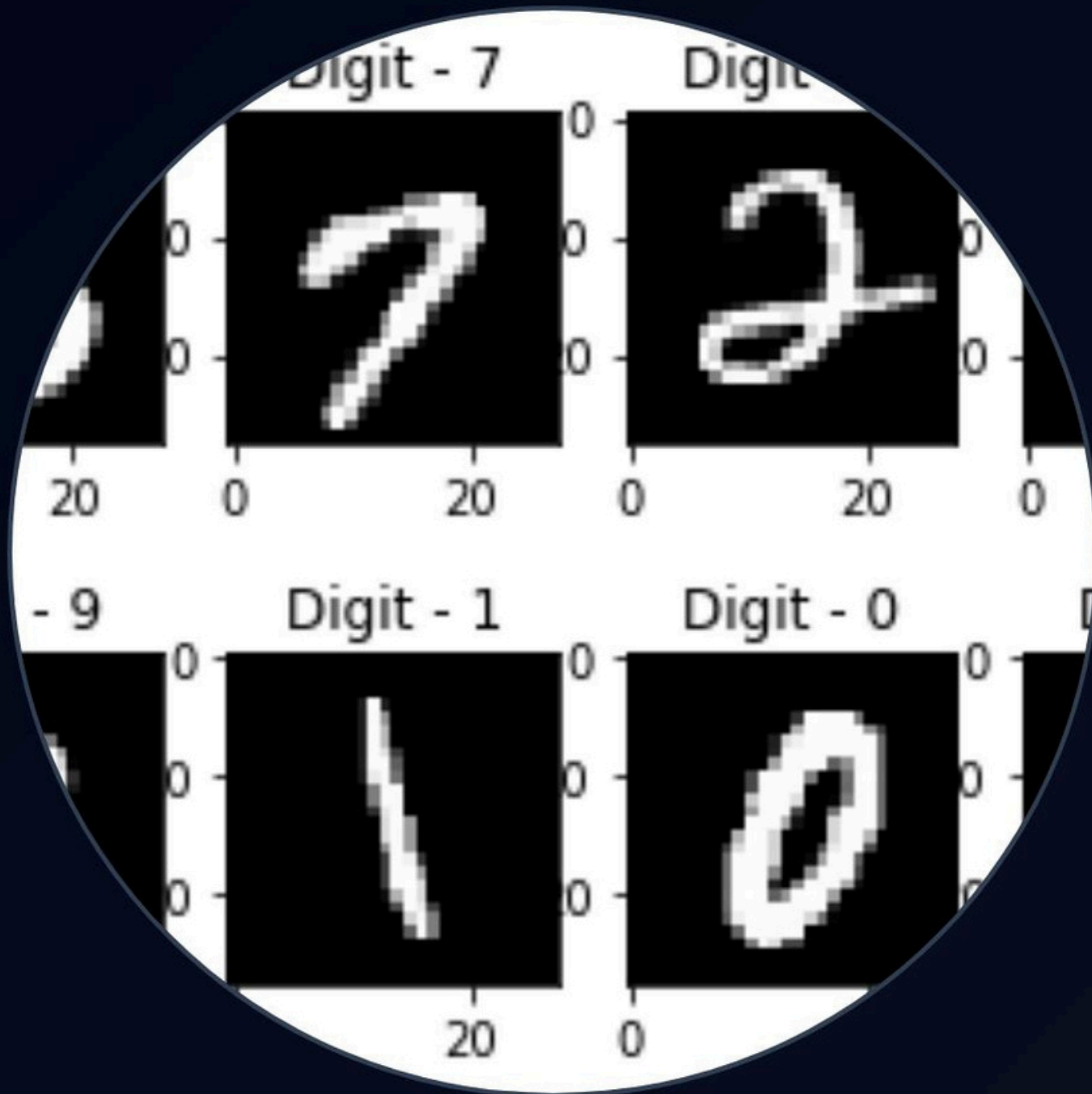
TensorFlow & Keras framework for building and training Sequential neural models.



Deployment

Streamlit web application interface for real-time digit recognition and visualization.

The MNIST Database



70,000 Grayscale Images

MNIST is the standard benchmark for evaluating image processing and computer vision algorithms.

- ✓ 60,000 Training Samples
- ✓ 10,000 Testing Samples
- ✓ Uniform 28x28 pixel dimensions

Data Preprocessing Pipeline

Preparing Raw Pixels

Images are normalized to accelerate the neural network's learning phase.

- **Normalization:** Scaling values to [0-1].
- **Flattening:** Vectorizing 2D pixel grids.
- **Inversion:** Enhancing custom image contrast.

Data processing visuals

Neural Network Architecture

Hidden Layers

Dense Layer: 128 Neurons with ReLU activation for complex feature extraction.

Dropout: 20% rate applied to prevent model overfitting.

Output Strategy

Softmax Layer: 10 Neurons corresponding to labels 0 through 9.

Outputs a probability score for each potential digit classification.

Training Performance

99%

Training Accuracy

High Precision Learning

The model reached peak accuracy using the Adam optimizer within only 5 training cycles (epochs).

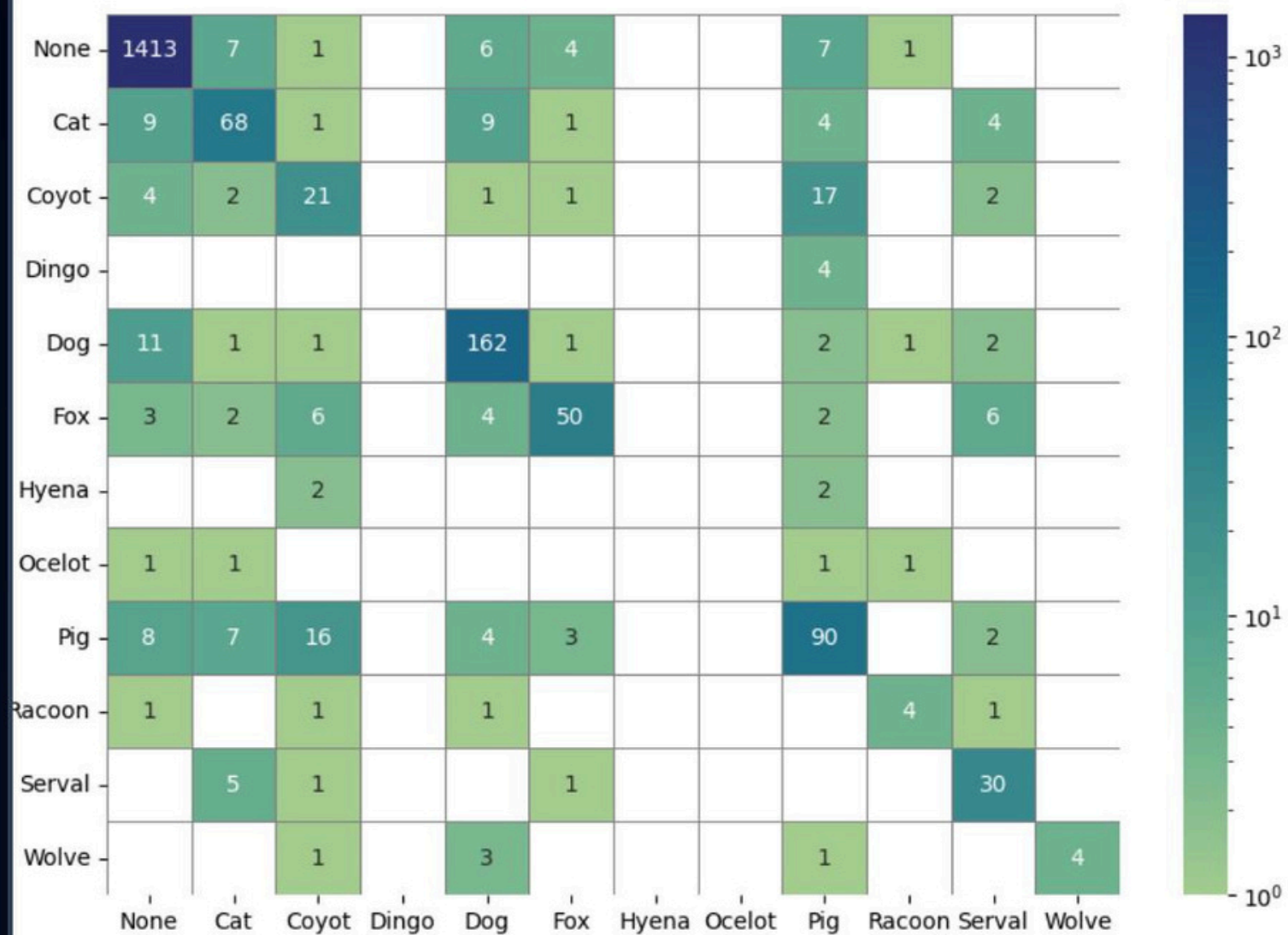
Val_Accuracy: 98.4% | Final Loss: 0.0425

Model Evaluation



High performance across training and unseen test datasets.

Error Analysis



Model Limitations

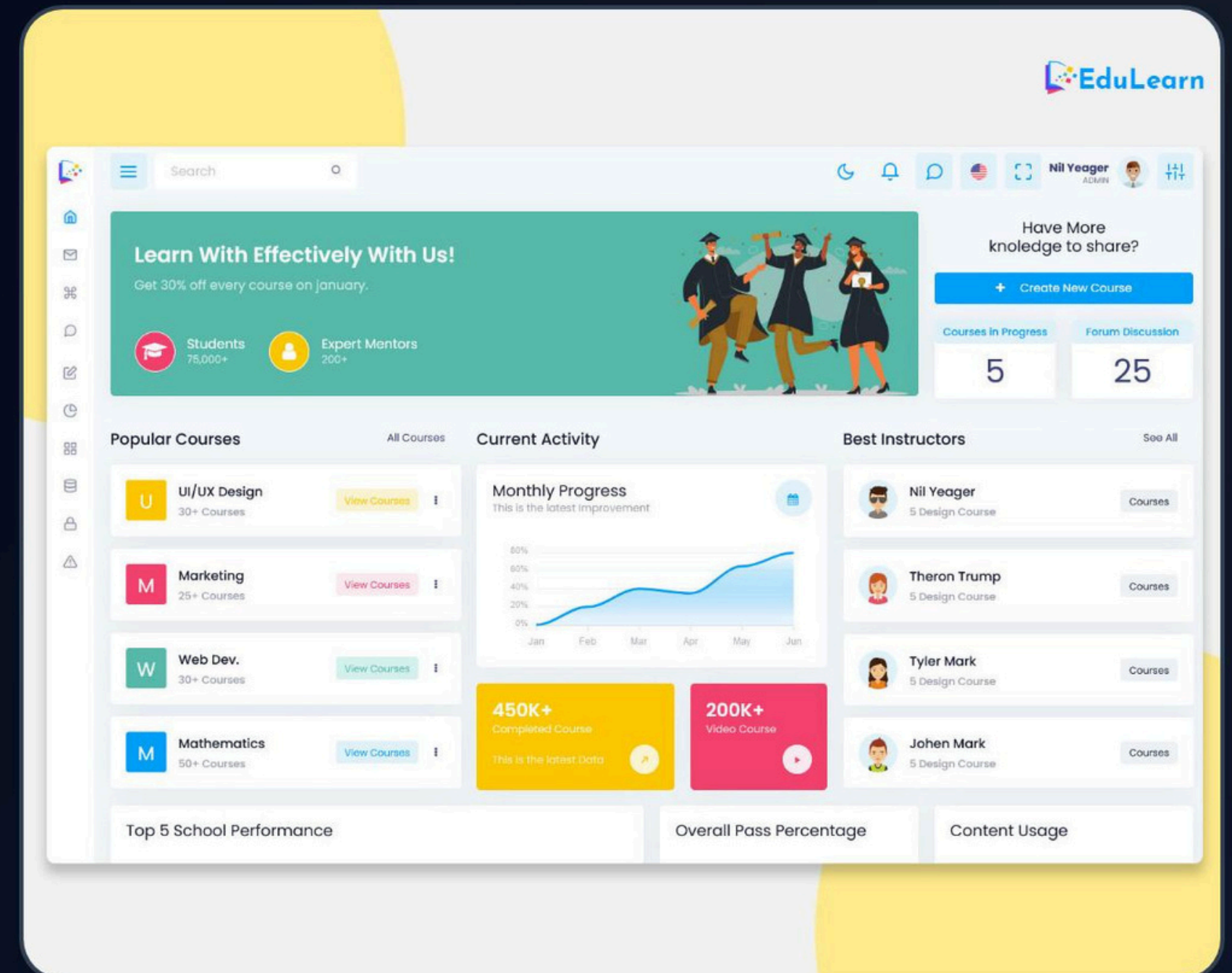
Minor errors occur between visually ambiguous digits like 4 and 9, or 3 and 8.

Analysis of the confusion matrix ensures model robustness and helps identify edge cases.

Streamlit Web Application

Deployment Interface

- Simple image upload.
- Confidence visualization.
- Instant prediction results.



Future Applications

-  **Logistics:** Sorting letters by hand-written zip codes.
-  **Finance:** Reading digit amounts on checks.
-  **Digitization:** Converting paper forms to digital data.
-  **Security:** Enhancing captchas and recognition.

Thank You!

Any Questions?